C28 Amines – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Structure and Basicity **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Primary amine				N bonded to one carbon
2	Secondary amine				N bonded to two carbons
3	Tertiary amine				N bonded to three carbons
4	Quaternary ammonium ion				N bonded to four carbons, positive charge
5	Why amines are bases				Lone pair on N accepts H^+
6	Why amines are nucleophiles				Lone pair on N
7	Order of base strength				Primary aliphatic amine > NH_3 > aromatic amine
8	Why aliphatic amines are stronger bases				Alkyl groups push electron density onto N
9	Why aromatic amines are weaker bases				Lone pair delocalised into ring
10	Use of quaternary ammonium salts				Cationic surfactants (fabric softeners)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C28 Amines – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Preparation and Reactions **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Amine from a halogenoalkane				Excess NH_3 in ethanol, heat, sealed tube
2	Amine from a nitrile				$LiAlH_4$ or H_2/Ni
3	Why the nitrile route is better				Gives only the primary amine
4	Reagents to reduce nitrobenzene				Sn + conc HCl, then NaOH
5	Equation for reducing nitrobenzene				$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$
6	Product of reducing nitrobenzene				Phenylamine
7	Use of aromatic amines				Making dyes
8	Amine + acid				Ammonium salt
9	Product of ethylamine + HCl				Ethylammonium chloride
10	Amine + acyl chloride				*N*-substituted amide

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